# CardioTwin: A Physiologically-Grounded Digital Twin for Early Warning of Hypertensive Crisis in the Indian Population

*Solution blueprint, revised and consolidated. React frontend, FastAPI backend, semi-synthetic data with documented provenance.*

## 1. Solution Overview

CardioTwin builds a patient-specific virtual cardiovascular system. It fuses static EHR data with dynamic wearable streams through a physiological model (a Windkessel arterial model) plus a machine-learning layer, rather than relying on a black-box classifier alone. The scope is deliberately narrow so every claim can be measured: one hero event, a crisis-range blood pressure episode predicted 6 and 12 hours ahead, plus rhythm irregularity detection from RR intervals as a supporting capability.

- **Hero event:** crisis-range BP episode (systolic ⩾ 180 and/or diastolic ⩾ 120 mmHg), predicted at 6-hour and 12-hour horizons.
- **Supporting capability:** AF-type rhythm irregularity detection from RR intervals, validated on real data. This is detection, not prediction.
- **Twin outputs on top:** Twin Drift Index, days-to-risk horizon, minimum-change prescriptions and an India environment layer.
- **Positioning:** a clinical decision-support proof of concept. It does not diagnose, and it never recommends starting, stopping or changing a drug.

### 1.1 Key USPs

1. **Physiological model plus ML hybrid.** A 3-element Windkessel arterial model with a simple heart-rate-dependent stroke volume term is personalized from the patient's EHR profile and combined with ML. The twin can answer what-if questions through the physiology, not only through correlations.
2. **India-calibrated risk scoring.** Scores built mostly on Western cohorts can misestimate risk in South Asians, who tend to develop CVD earlier, at lower BMI, with higher Lp(a) and a high metabolic syndrome burden. CardioTwin adjusts a base score using published Indian data (ICMR-INDIAB, INTERHEART South Asia, CARRS, Jaipur Heart Watch). Every adjustment is cited or labeled an assumption.

3. **Dual-timescale architecture.** A Slow Twin (EHR-driven, chronic risk) and a Fast Twin (wearable-driven, acute state) are combined by a Fusion layer, so the same acute pattern lands in a higher alert tier for a high-chronic-risk patient than for a low-risk one.
4. **Explainable mechanistic pathways.** Every alert shows a pathway such as: missed doses, rising peripheral resistance, sympathetic activation (HRV falls, LF/HF rises), simulated BP reaching 180/120. The pathway is a rule-based template filled with SHAP and Windkessel values. It explains the model; it is not causal inference.
5. **Semi-synthetic data with documented provenance.** The cohort is built from real open recordings and published statistics, and a datasheet records whether each field is real-fitted, derived or assumed.
6. **Twin Drift Index.** One 0-100 score showing how far the patient's current physiology has drifted from their own baseline (resting HR, HRV, sleep, activity). Doctors get one trend line instead of five charts. Pitch: the patient's vital drift, not just their risk.
7. **Minimum-change prescriptions.** Instead of guessing slider values, the twin searches for the smallest lifestyle or adherence change that moves risk down one tier, for example '+2,500 steps a day and sleeping 40 minutes earlier moves risk from High to Elevated'. It is a model-based estimate under our simulation assumptions, not causal proof.
8. **India environment layer.** AQI, heat waves and Diwali-season smog are plausible cardiac triggers in India and are rarely modeled. AQI, temperature and festival or fasting flags enter as time-varying features. Effect sizes are used only when a teammate has verified a source; otherwise they are shown as modeled assumptions.
9. **Days-to-risk horizon.** Instead of only a probability, the twin shows the estimated time before risk crosses the High tier if the current trend continues, so the doctor sees how urgent the situation is as well as how bad.

### 1.2 Non-goals (kept out of the 3-day build)

MI risk escalation, AF episode prediction, Cox survival modeling, a multi-task LSTM, inferred medication non-adherence, MIMIC-IV, Synthea, and cloud deployment. Each is listed in section 12 with the reason it was cut.

## 2. Clinical Scope

**Target patient:** an Indian adult aged 45-70 with established risk factors (hypertension, dyslipidemia, diabetes, family history), monitored after diagnosis or a cardiac event. The cohort samples ages 35-80 so the models see variety, and 45-70 is the primary evaluation

subgroup.

## 2.1 Hero event definition

- **Event:** a crisis-range BP episode, defined for this PoC as at least two consecutive BP readings within 60 minutes with systolic ⩾ 180 or diastolic ⩾ 120 mmHg.
- **Naming:** clinically, a hypertensive crisis is split into urgency and emergency by the presence of organ damage. Our data cannot show organ damage, so slides and the README say 'crisis-range BP episode'.
- **Prediction target:** at each 5-minute update, the label is 1 if an episode starts within the next 6 hours (and, separately, within the next 12 hours). Windows already inside an episode are excluded.
- **Inputs:** HR and HRV, BP readings, activity, sleep, medication timing and adherence, environment features and the EHR vector.

## 2.2 Signals and sampling

| Signal | Sampling in the PoC | Realism note |
|---|---|---|
| HR and HRV (SDNN, RMSSD, LF/HF) | 5-minute windows computed from RR intervals | Assumes a chest strap or good-quality PPG; consumer HRV quality varies |
| Blood pressure | Every 30 min awake, every 60 min asleep (ambulatory-monitor style) | Assumption: an ambulatory or cuffless sensor. Consumer wearables do not give true continuous BP, and cuffless accuracy is a stated limitation |
| Steps and activity | 5-minute counts | Typical wearable output |
| Sleep | Nightly: duration, efficiency, bedtime, regularity | Stage percentages are optional |
| Environment | Daily per city: AQI, PM2.5, temperature, festival flag | Synthetic series shaped like Indian seasons |
| EHR | Static fields plus labs at 1-3 time points | Seeded from open datasets and published statistics |

### 2.3 Supporting capability: rhythm irregularity detection

Flag irregular rhythm from RR-interval statistics over short windows (for example coefficient of variation, RMSSD relative to mean RR, Shannon and sample entropy, Poincaré SD1/SD2). Train and test on the real MIT-BIH Atrial Fibrillation Database with record-wise splits. This is detection only, because real data can validate a detector but cannot support an AF forecast we could defend.

### 2.4 Outputs on every prediction

- **Days-to-risk:** estimated time before risk crosses the High tier if the current trend continues.
- **Twin Drift Index:** a 0-100 deviation-from-baseline trend. In our simulated episodes it moves before BP does, but that is built into the generator. It demonstrates the mechanism and is not evidence that real patients behave this way.
- **Minimum-change prescription:** for any patient at Elevated risk or above, the smallest lifestyle or adherence change that lowers the tier, as a model-based estimate.
- **Environmental context:** AQI, temperature and festival flags in the explanation.

## 3. Dataset Strategy

Data is where this project is won or lost. The plan is a semi-synthetic cohort: real open recordings and published statistics seed every field, synthetic generation fills only what no open dataset provides (outpatient BP and crisis episodes), and a datasheet documents the provenance of every field. The cohort is never described as real patient data.

### 3.1 EHR seeding from real open data

**Principle:** fit marginal distributions and correlations from real open datasets, shift them with published Indian statistics, and record where every value came from. The teammate confirms licence and access for each source on day 1.

- **UCI Heart Disease (Cleveland) and Heart Failure Clinical Records:** age, sex, BP, cholesterol and comorbidity structure.
- **NHANES:** correlations between lipids, BP, HbA1c, BMI and age.
- **Published Indian statistics:** shifts for BMI, waist circumference, smoking, tobacco chewing, diabetes and hypertension control, each cited. Where no source exists, the value is labeled an assumption.

- **Not used in this build:** MIMIC-IV (credentialing delay, hospital-based population, and its data terms generally do not allow republishing derived data in a public repo) and Synthea (heavy customization needed to say anything about India).

### 3.2 Cohort design (1,000 patients, 30 days each)

1,000 patients keeps generation, training and plotting fast. Scale to 5,000 only if the pipeline is already stable. Thirty days gives a 14-28 day personal baseline plus a held-out period; the live demo replays a 7-day window.

**Demographics and lifestyle** (fitted from real data where possible, then shifted with published Indian statistics; every value marked sourced or assumed in the datasheet):

| Field | Calibration |
|---|---|
| Age | Sampled 35-80, shaped to published CARRS summary statistics |
| Sex | About 60% male, 40% female (assumption) |
| BMI | Mean about 24.5; overweight above 23 and obese above 25 (WHO Asia-Pacific cutoffs) |
| Waist circumference | Risk cutoffs: men above 90 cm, women above 80 cm |
| Smoking | About 28% current among men, 2% among women (GATS India, to verify) |
| Tobacco chewing | About 18% (to verify), a distinctly South Asian risk factor |
| Physical activity | About 54% sedentary (ICMR-INDIAB, to verify) |
| Diet | About 30% vegetarian (to verify) |
| Family history | Premature CVD (before 55 in men, before 65 in women) |
| City and socioeconomic status | Drive the environment feed, adherence and regional prevalence (assumption) |

**Medical history (core):** hypertension with duration, type 2 diabetes, dyslipidemia, prior MI, stroke, PCI/CABG, AF history, CKD stage and recent hospitalization. *Optional only if cheap:* COPD, hypothyroidism, rheumatic heart disease.

**Labs (core):** HbA1c, fasting glucose, total/LDL/HDL cholesterol, triglycerides, creatinine/eGFR and potassium, sampled at 1-3 time points in the six months before day 0, not as full time series. *Optional:* CRP, homocysteine, Lp(a), BNP. Fields with no seed dataset are marked 'assumed'. Troponin and TSH are dropped because they do not feed the hero event.

**Medications (core):** ACE inhibitor (ramipril, enalapril), ARB (telmisartan, losartan), beta-blocker (metoprolol, atenolol), calcium channel blocker (amlodipine), thiazide-type diuretic (chlorthalidone, hydrochlorothiazide), statin (atorvastatin, rosuvastatin), aspirin and metformin. Baseline adherence per patient is 50-85% depending on class (assumption), with day-level missed-dose patterns. Antiplatelets beyond aspirin, anticoagulants and insulin are optional and unused by the BP model.

## 3.3 Wearable data: real baselines plus synthetic events

**Real sources used as baselines and for validation:**

- **MMASH (PhysioNet):** RR intervals, activity and sleep from healthy adults, resampled as each patient's HR, HRV, sleep and activity baseline.
- **MIT-BIH Normal Sinus Rhythm Database:** clean RR-interval baselines.
- **PPG-DaLiA and the open Fitbit dataset on Kaggle:** heart rate and activity patterns in daily life.
- **MIT-BIH Atrial Fibrillation Database (PhysioNet):** real RR intervals for validating rhythm irregularity detection.

**Be honest about seed size.** Each real wearable source has only tens of subjects. The cohort is described as 'resampled and perturbed from N real recordings', never as N independent real people.

**Semi-synthetic generator.** Resample real segments as each patient's baseline, then layer effects on top. BP and crisis episodes stay synthetic because no open dataset links outpatient BP to an EHR; they are built from published ambulatory BP physiology.

- Disease and medication effects (age, diabetes, hypertension, prior MI, beta-blocker) applied as shifts to the real resting HR and HRV baseline.
- Ambulatory BP physiology: nocturnal dip, morning surge, and antihypertensive effect scaled by adherence.

- A crisis prodrome in the hours before an episode: rising HR and BP, falling HRV, rising LF/HF, with onset time and strength varying per patient.
- An Indian-lifestyle activity pattern (morning and evening walks, sedentary work hours), scaled down for sedentary patients.
- Environment-triggered episodes (several days of Very Poor AQI plus short sleep) and missed-dose episodes alongside the baseline prodrome.
- Realistic noise, missing data and device dropouts.
- **Prevalence:** about 10-15% of patients get at least one episode in 30 days. This is enriched on purpose compared with real outpatients, so every AUPRC is reported next to the prevalence.
- **Honest caveat:** the models learn effects the generator encodes. Results show that the pipeline works on a controlled simulation, not that the effects exist in real patients.

## 3.4 Dataset organization

Five groups: **EHR data** (demographics, history, labs, medications), **wearable data** (HR, HRV, BP, activity and sleep per patient as Parquet), **derived features** (temporal features, drift components, physiological outputs), **event labels** (crisis-range episodes and detected rhythm irregularity) and **metadata** (datasheet, data dictionary, India calibration parameters, generation config and random seed). Structured data in SQLite, time series in Parquet.

## 3.5 India environment and calendar data

A daily environment feed per patient city. Fields: AQI (CPCB categories: Good, Satisfactory, Moderate, Poor, Very Poor, Severe), PM2.5, temperature, humidity, a heat-wave flag and a festival or fasting flag (Diwali window, Holi, Navratri, Ramadan fasting, winter smog season).

- **Synthetic series:** shaped like Indian city seasons: winter smog peaks in northern cities such as Delhi-NCR, pre-monsoon heat in central and western cities, cleaner monsoon months and a short AQI spike around Diwali. Validate shapes against open CPCB data where it can be downloaded, and describe the series as synthetic.
- **Linking:** each patient is assigned a city, so they see the same environment across the whole simulation.
- **Teammate task:** verify the published evidence linking air pollution, heat and festival-season changes to cardiovascular events before any effect size is used.

### 3.6 Realism checks and datasheet

Three checks go into the README and the video:

- Plots comparing synthetic and real distributions (histograms and a correlation heatmap).
- A train-on-synthetic, test-on-real check for the **EHR branch only**, using the fields shared with UCI Cleveland and predicting heart disease presence there (it does not test crisis prediction). The rhythm detector is trained and tested on real records, so it has no synthetic-to-real step.
- A **datasheet** with one row per field: field name, status (real-fitted, derived or assumed), source and citation or the stated assumption, and notes.

## 4. Technical Architecture

### 4.1 Layers

- **Data layer:** EHR cohort seeded from open datasets, the semi-synthetic wearable generator seeded from real recordings, and the environment feed.
- **Feature layer:** temporal and physiological features, personal baselines and the Twin Drift Index.
- **Digital Twin engine:** Slow Twin (chronic risk), Fast Twin (crisis model and rhythm detector), Windkessel physiological model and fusion.
- **Alert and insight engine:** calibrated risk, alert policy, explainability, days-to-risk and minimum-change prescriptions.
- **Doctor-facing dashboard:** a React app on a FastAPI backend with WebSocket replay.

The environment feed joins the data layer, the Drift Index runs in the feature layer, and the days-to-risk estimator and prescription search sit on top of the fusion output. All are exposed through FastAPI endpoints and shown as React panels.

### 4.2 Technology stack

| Component | Technology | Why |
|---|---|---|
| Backend | FastAPI, Uvicorn, Pydantic | Async, OpenAPI docs, typed schemas, good fit for serving models |

| Component | Technology | Why |
|---|---|---|
| Streaming | FastAPI WebSockets | Replay pre-generated data as a live stream |
| Data | Pandas, NumPy, Parquet, SQLite | No server to run |
| ML | LightGBM, scikit-learn; optional PyTorch GRU | Strong tabular baseline first; GRU only if time allows |
| Physiology | SciPy | Windkessel simulation and optional bounded calibration |
| Explainability | SHAP | Feature contributions per alert |
| Frontend | React (Vite, TypeScript), Tailwind CSS, shadcn/ui | Fast, clean, medical-grade look |
| Charts | Plotly.js or Recharts | Gauges, time series, Sankey pathway |
| Data fetching | TanStack Query plus a light state store | Caching, polling, live state |
| Delivery | Local run with a one-command README | Deployment is future scope |
| Version control | Git and GitHub | Required for submission |

## 4.3 API endpoints

| Endpoint | Purpose |
|---|---|
| GET /patients | Demo patients with risk summary |
| GET /patients/{id} | Profile: demographics, history, medications, latest labs |
| GET /patients/{id}/wearable | Wearable time series for a window |
| GET /patients/{id}/twin | Twin state: fused risk, chronic and acute risk, physiological parameters |
| GET /patients/{id}/predictions | Crisis probabilities at 6 h and 12 h, ETA, predicted BP trajectory |

| Endpoint | Purpose |
|---|---|
| GET /patients/{id}/rhythm | Rhythm irregularity flag and supporting RR statistics |
| GET /patients/{id}/explanation | Narrative, SHAP contributions, pathway data |
| POST /patients/{id}/scenarios | What-if run (missed dose, exercise, stress) |
| POST /patients/{id}/prescriptions | Minimum-change search; ranked options |
| GET /patients/{id}/drift | Drift Index trend with per-signal contributions |
| GET /patients/{id}/horizon | Estimated time to High tier, with range |
| GET /patients/{id}/environment | AQI, temperature and festival flags on the patient timeline |
| GET /patients/{id}/alerts | Alert history |
| GET /patients/{id}/labs | Lab values |
| WS /ws/patients/{id} | Live twin updates and alerts (replay) |
| POST /simulate/playback | Play, pause and speed |
| GET /datasheet | Data provenance table, shown in the dashboard |
| GET /health | Service and model status |

## 4.4 Backend notes

- Project structure: routers (patients, twin, predictions, scenarios, alerts, stream), services (slow twin, fast twin, physiology, fusion, explainability, India calibration), Pydantic schemas, data access and a models folder loaded once at startup.
- Agree the Pydantic schemas and OpenAPI contract first, and generate mock JSON from them so the React app can be built before the models exist.
- Enable CORS for the React origin.
- A simulation service replays pre-generated 7-day wearable data in accelerated time over WebSockets. This makes the live demo reliable without a device.
- Cache Windkessel results per patient.